Nucleic Acids
Lecturer 5

Presented By

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Contents

Chapter III
DNA

Chapter Three: DNA

- 1- DNA base composition
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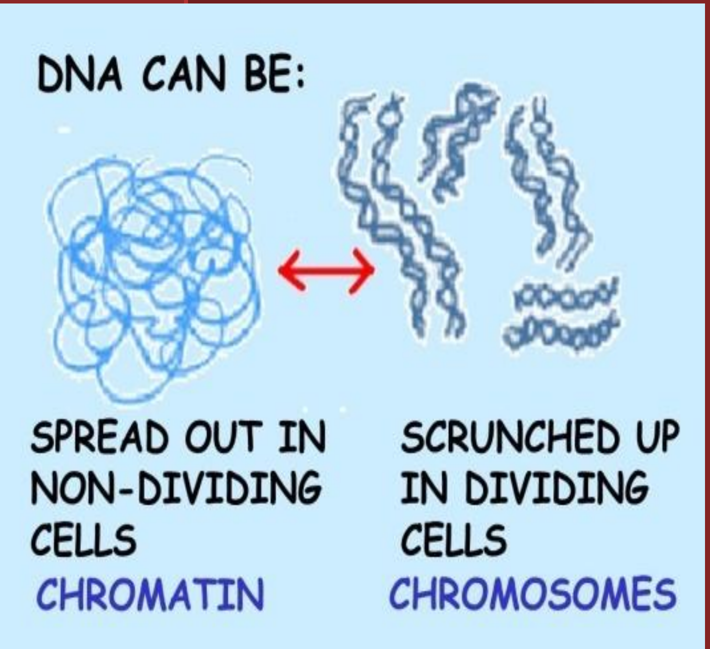
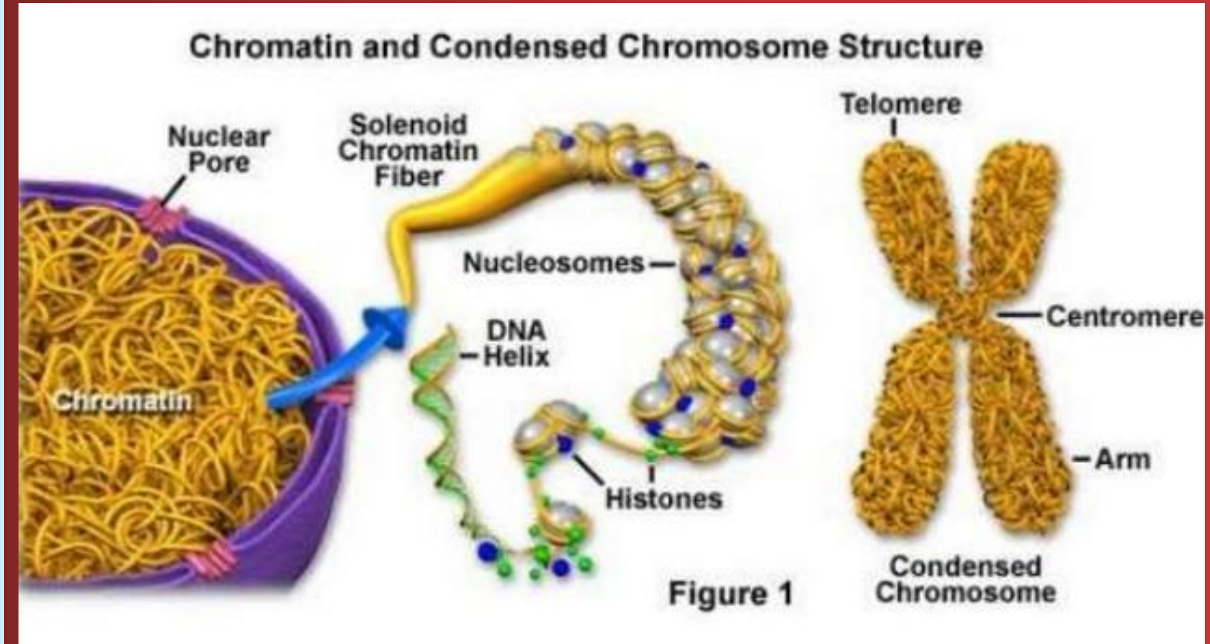
- 3- A, B, and Z forms of DNA
- 4- Packing DNA into chromosomes

4- Packing DNA into chromosomes

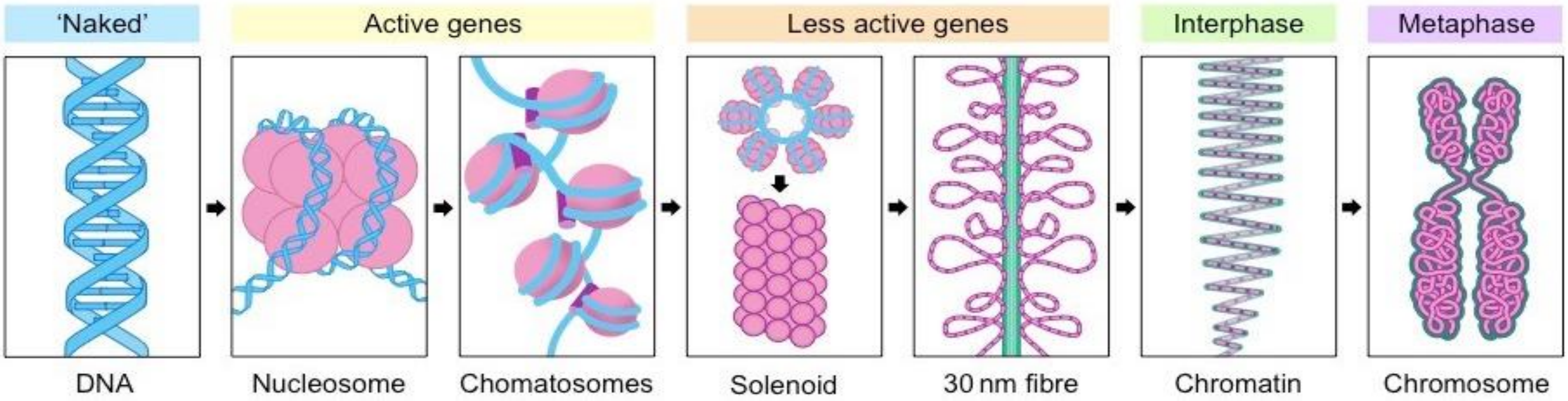
- **a- Eukaryotic chromosomes:** The nucleus of each cell in our bodies contains approximately 1.8 meters of DNA in total, although each strand is less than one millionth of a centimeter thick. This DNA is tightly packed into structures called chromosomes, which consist of long chains of DNA and associated proteins.

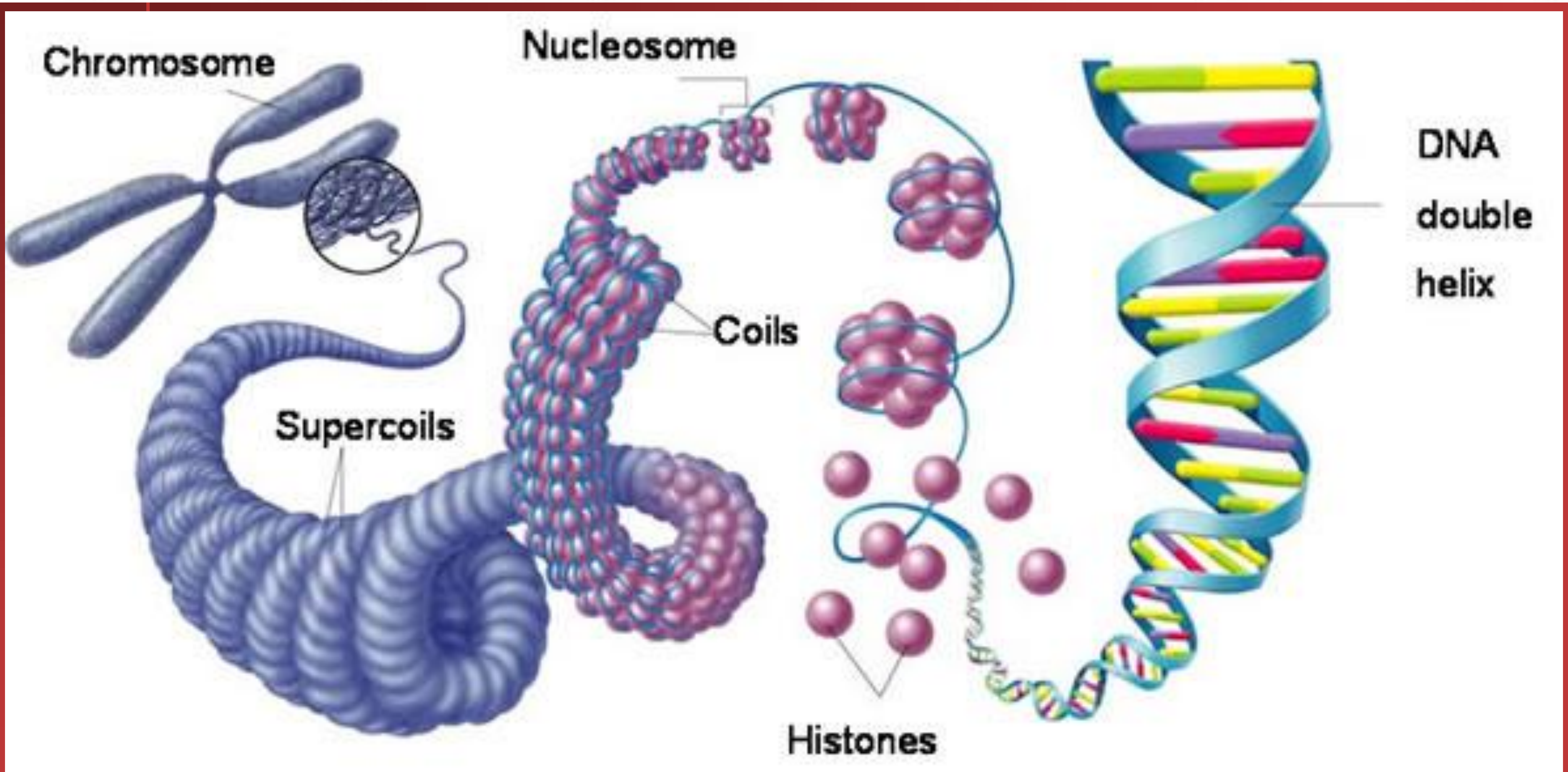




4- Steps of Packing DNA into chromosomes

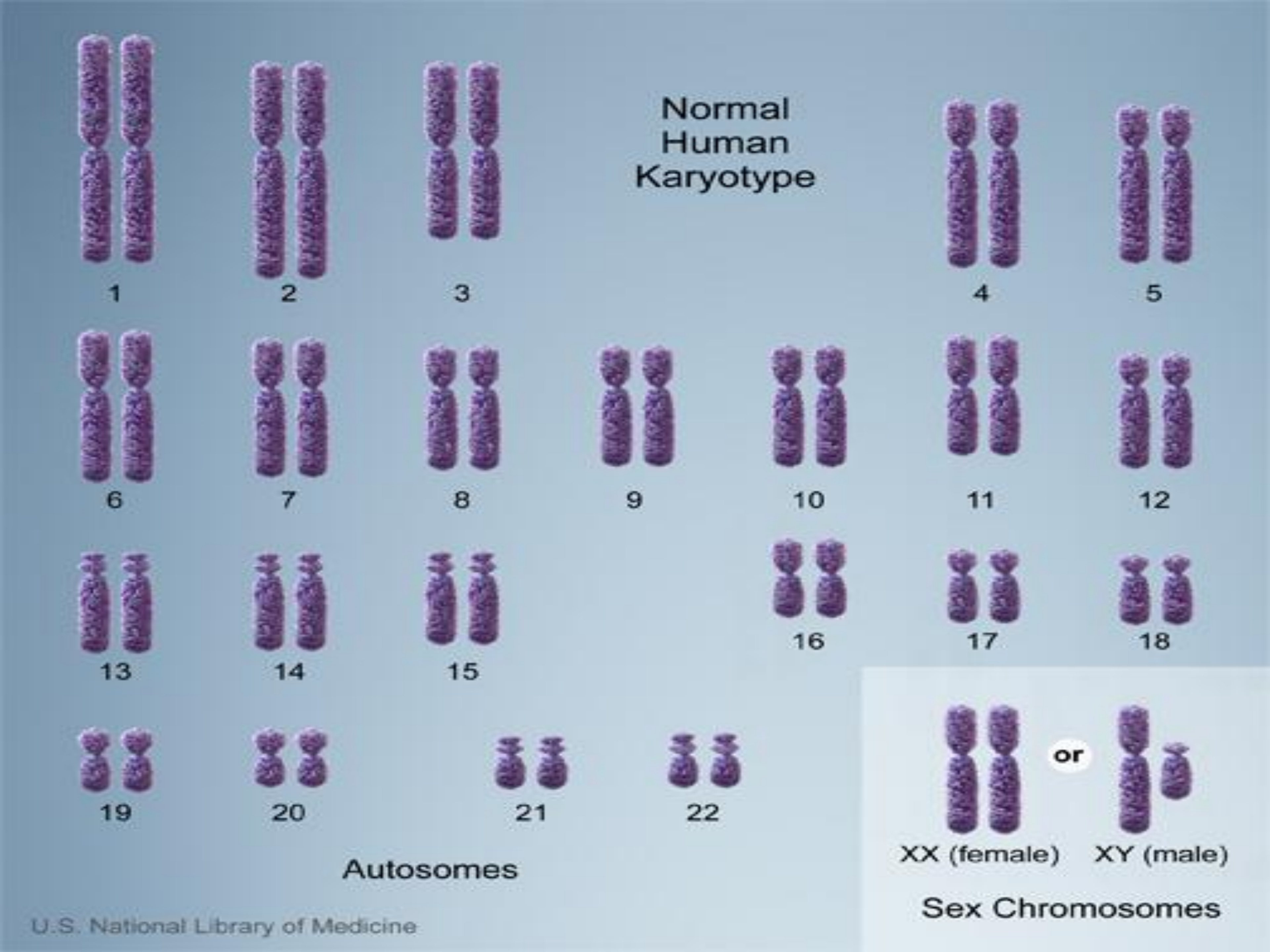
- DNA molecules are tightly wound around proteins - called **histone** proteins. A DNA strand of 150 to 200 nucleotides long is wrapped twice around a core of eight histone proteins to form a structure called a **nucleosome**.
- The chains of histones are coiled in turn to form a **solenoid**, Further coiling of the solenoids forms the structure of the chromosome proper.
- Each chromosome has a p arm and a q arm. The p arm (from the French word 'petit', meaning small) is the short arm, and the q arm (the next letter in the alphabet) is the long arm.
- In their replicated form, each chromosome consists of two **chromatids**. The chromosomes - and the DNA they contain - are copied as part of the cell cycle, and passed to daughter cells through the processes of mitosis and meiosis.


'Naked'
Active genes
Less active genes
Interphase
Metaphase
DNA
Nucleosome
Chomatosomes
Solenoid
30 nm fibre
Chromatin
Chromosome


Chromosome
Nucleosome
DNA
double
helix
Coils
Supercoils
Histones

Human beings chromosomes

- Human beings have 46 chromosomes, consisting of 22 pairs of **autosomes** and a pair of **sex chromosomes**:
- Two X sex chromosomes for females (XX) and an X and Y sex chromosome for males (XY).
- One member of each pair of chromosomes comes from the mother (through the egg cell); one member of each pair comes from the father (through the sperm cell).
- A photograph of the chromosomes in a cell is known as a **karyotype**. The autosomes are numbered 1-22 in decreasing size order.



Normal
Human
Karyotype
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
Autosomes
or
XX (female)
XY (male)
Sex Chromosomes
U.S. National Library of Medicine

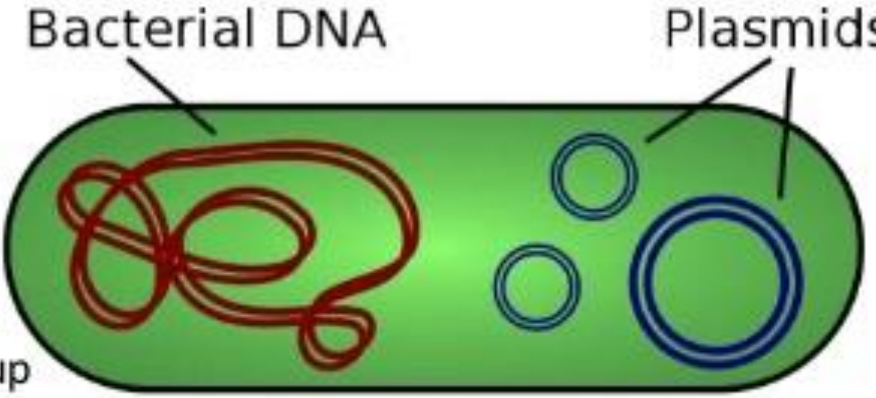
b- Prokaryotic Chromosomes:

- Most bacteria contain a single, circular chromosome. (There are exceptions: some bacteria - for example, the genus Streptomyces - possess linear chromosomes).
- The chromosome - together with ribosomes and proteins associated with gene expression - is located in a region of the cell cytoplasm known as the nucleoid. The circular chromosome of the bacterium Escherichia coli consists of a DNA molecule approximately 4.6 million nucleotides long.
- In addition to the main chromosome, bacteria are also characterized by the presence of extra-chromosomal genetic elements called plasmids.
- These relatively small circular DNA molecules usually contain genes that are not essential to growth or reproduction.

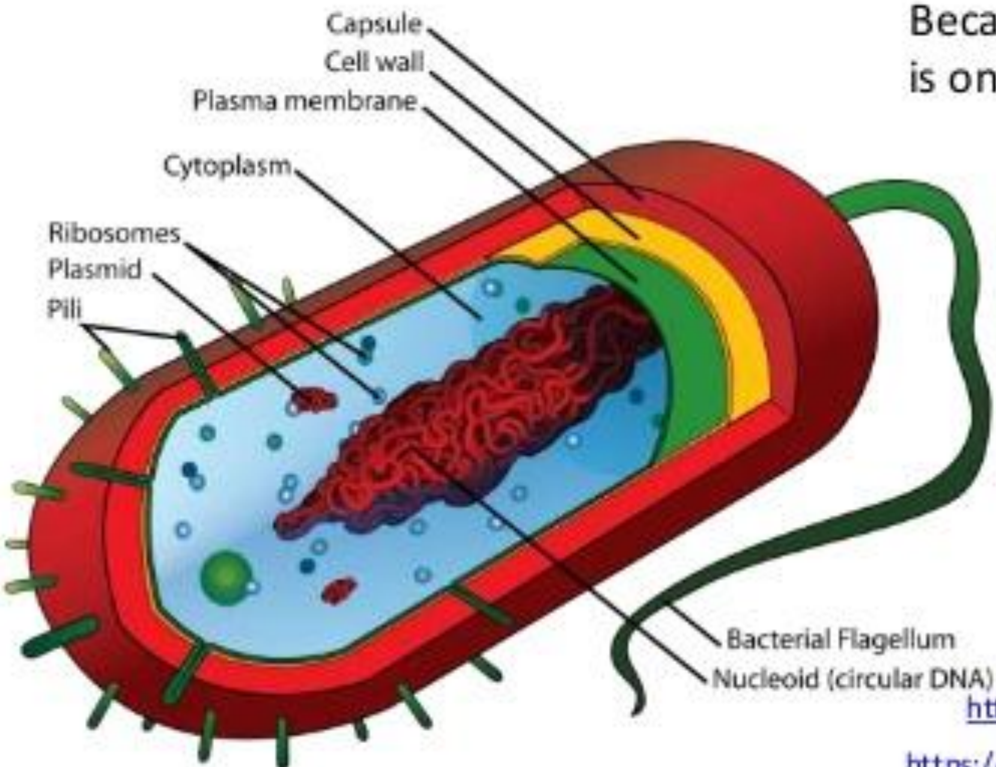
3.2.U1 Prokaryotes have one chromosome consisting of a circular DNA molecule.
3.2.U2 Some prokaryotes also have plasmids but eukaryotes do not.

Prokaryotes have two types of **DNA**:

- **single chromosome**
- **plasmids**



The single prokaryotic chromosome is coiled up and concentrated in the nucleoid region.



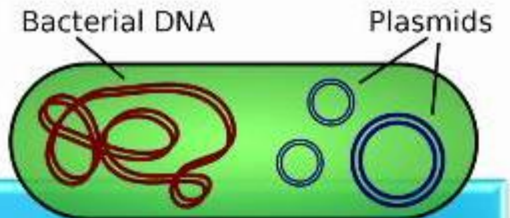
Because there is only a single chromosome there is only one copy of each gene.

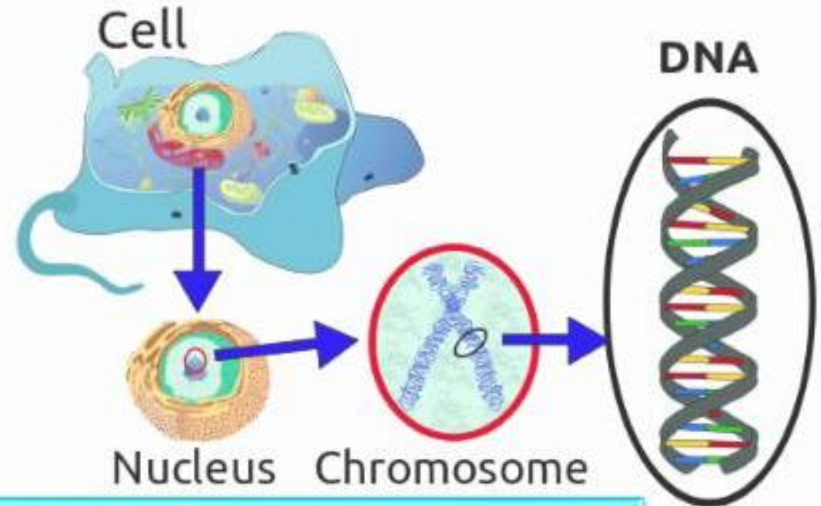
A copy of the chromosome is made just before cell division (by binary fission).

https://commons.wikimedia.org/wiki/File:Plasmid_%28english%29.svg

https://commons.wikimedia.org/wiki/File:Average_prokaryote_cell-_en.svg

PROKARYOTIC VS. EUKARYOTIC CHROMOSOMES





Prokaryote

- a circular DNA molecule
- naked - no associated proteins
- plasmids often present
- one chromosome only

Eukaryote

- Contains a linear DNA molecule
- associated with histone proteins
- no plasmids
- two or more different chromosomes